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TRANSISTOR AMPLIFIERS

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This **Basic Electronics Tutorials eBook** is focused on bipolar transistor amplifiers with the information presented within this ebook provided “as-is” for general information purposes only.

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1. TRANSISTOR AMPLIFIERS INTRODUCTION

Amplifiers are everywhere and are the basic building block for a vast majority of electronic circuits. Electronic amplifiers are much more than just a circuit for the reproduction of audio and music applications. Most electronic circuits or systems require some form of amplification of low voltage or current signals, and as such there are many different types of electronic circuits available which can be classed as amplifiers.

A linear amplifier is generally a circuit which receives an input signal and produces a corresponding output signal that is a fairly good replica of the input signal while at the same time increasing the amplitude of the signal. But not all amplifier circuits are the same as they can be classified according to their circuit configurations and the type of signals they must amplify, large or small.

Since signals can be transmitted as either a voltage or as a current, electronic amplifier circuits include Operational Amplifiers, Instrumentation Amplifiers, Small Signal, Large Signal and Power Amplifiers.

An amplifier is an electronic device which can increase the voltage, current, or power of its input signal

The classification of an amplifier circuit depends upon the size of the input signal, large or small, its physical configuration and how it processes the input signal. That is the relationship between input signal and output load.

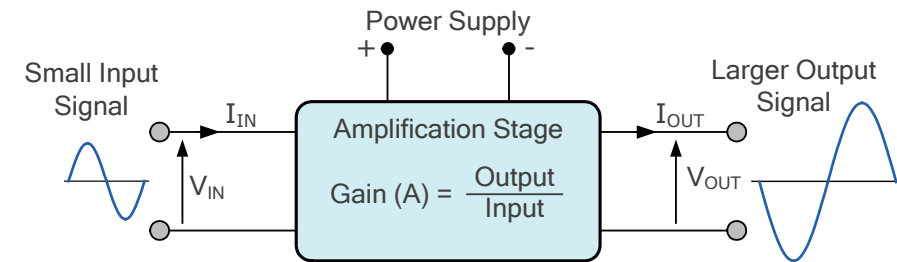
At the most basic level, there are commonly four categories of amplifier classified according to their input and output signal (voltage or current) and the relationship it has between its input and output.

- **Voltage Amplifier:** These are the vast majority of amplifier which amplify a given input voltage for a larger output voltage, ($A_v = V_{OUT}/V_{IN}$). Voltage amplifiers are characterised as having a high input impedance and a low output impedance.
- **Transconductance Amplifier:** This is an amplifier that changes its output load current according to changes in its input voltage allowing voltage control of its gain, ($A = I_{OUT}/V_{IN}$).

- **Current Amplifier:** This is an amplifier that amplifies a given input current to a much higher output current, ($A_i = I_{OUT}/I_{IN}$). It is characterized by a low input impedance and high output impedance.
- **Transimpedance Amplifier:** This is an amplifier that changes output voltage according to changing input current, ($A = V_{OUT}/I_{IN}$). It is also known as a summing amplifiers or as current-to-voltage converters.

In all their different forms, amplifiers can be thought of as a simple box or block containing the amplifying device, such as a Bipolar Transistor, Field Effect Transistor or Operational Amplifier. It has two input terminals and two output terminals (ground being common) with the output signal being much greater than that of the input signal as it has been “Amplified”.

FIGURE 1. BASIC AMPLIFIER MODEL



This simple two-port model of an amplifier can be used to find the relationship between the input and output signals to determine various parameters, such as, input impedance, output impedance, voltage or current gain as well as the DC set point and operating parameters of the amplifier.

2. AMPLIFIER GAIN

For an amplifier to provide “amplification”, it must be able to accept an input signal and produce a corresponding output signal which is greater in amplitude than the original input signal. The amplified difference between the input and output signals is known as the **Gain (A)** of the amplifier.

Gain is a measure of how much an amplifier “amplifies” its input signal. For example, if we have an input signal of 1 volt and an output of 50 volts, then the gain of the amplifier would be “50”. In other words, the input signal has been increased by a factor of 50.

In Electronics gain is commonly given the symbol “**A**”, for Amplification. Amplifier gain is simply the ratio of the output divided-by the input. That is the “output signal divided by the input signal”. Gain has no units because it is a ratio.

There are three different kinds of amplifier gain. These are: Voltage Gain (A_V) for voltage amplifiers, Current Gain (A_I) for current amplifiers, and Power Gain (A_P) for power amplifiers.

$$\text{Voltage Gain, } A_V = \frac{\text{Output Voltage}}{\text{Input Voltage}} = \frac{V_{OUT}}{V_{IN}}$$

$$\text{Current Gain, } A_I = \frac{\text{Output Current}}{\text{Input Current}} = \frac{I_{OUT}}{I_{IN}}$$

$$\text{Power Gain, } A_P = \frac{\text{Output Power}}{\text{Input Power}} = \frac{P_{OUT}}{P_{IN}}$$

Remember that Gain (**A**) has no units as it is a ratio between output and input. The power gain (A_P) of an amplifier can also be calculated from the product of voltage gain and current gain as: $A_P = (A_V \times A_I)$.

Generally, a single amplifier stage which uses just one amplifying device is not sufficient to supply the required amount of amplification, especially when very small input signals are involved. Therefore, it is necessary to use two or more amplifier stages in cascade so that the output from the first stage is fed directly into the second stage, and so on.

When cascading together amplifier circuits, the overall gain of the circuit is equal to the product of each stage. For example, the gain of the first stage may be 10 and the gain of the second stage may be 50. Then the overall gain would be 500, ($500 = A_1 \times A_2 = 10 \times 50$).

Similarly, the overall gain for an amplifier circuit consisting of 3-stages together would therefore be: $A = A_1 \times A_2 \times A_3$ and so on for any number of amplifier stages.

The problem of expressing amplifier gain in this way is that the overall gain of multi-stage amplifier circuits can quickly become a large number. It is more common to express the overall gain in a logarithmic unit called a **decibel (dB)**.

The advantage of expressing the gain in decibels is that successive gains are simply added together. That is they sum together. The decibel (dB) is a Base-10 logarithmic unit of power change.

Amplifier Gain is commonly expressed in decibel units

$$dB_{(POWER)} = 10 \cdot \log_{10} [P_{OUT} / P_{IN}]$$

Thus our example above of an overall gain of 500 would be represented as:

$$10 \cdot \log_{10}(10) + 10 \cdot \log_{10}(50) = 10 + 17 = 27dB$$

According to Ohms Law, power can also be represented in terms of voltage and current. Then the decibels of voltage and current are given as:

$$dB_{(VOLTAGE)} = 20 \cdot \log_{10} [V_{OUT} / V_{IN}]$$

$$dB_{(CURRENT)} = 20 \cdot \log_{10} [I_{OUT} / I_{IN}]$$

The only difference when defining the power, voltage, or current as a decibel (dB) value is the constant of **10** or **20**. For the dB ratio to be correct in all instances the two quantities must both have the same units, either watts, milliwatts, volts, millivolts, amperes or milliamperes, or any other standard unit.

Note also, that if the decibel gains are negative numbers, for example, -3dB. Then this is considered a loss. Practical multi-stage amplifiers with cascaded sections can have both positive gains and negative gains present in the same circuit. The overall gain would still be the sum of the individual gains expressed in decibels.

3. TRANSISTOR AMPLIFIER CONFIGURATIONS

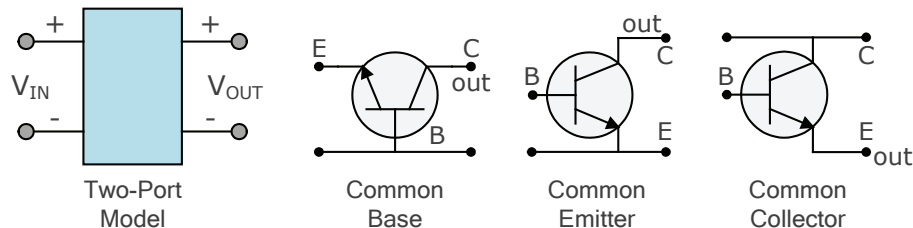
Bipolar Junction Transistors, or BJT's can be used either as amplifiers, or as electronic switches. In switching applications, the bipolar transistor is operated between the cut-off (off), and the saturation (on) region of its characteristic operating curves. Whereas when the transistor is used as an amplifier producing an output signal with more power than the input signal, it is said to be operating in its active mode.

Since a bipolar transistor is a 3-terminal device, when used as an amplifying device the input signal is applied between two transistor terminals, while the output signal is taken from between another two terminals. Since there are only three terminals to a transistor, one of the terminals must be common to both the input and output.

Then depending on which of the three transistor terminals is common between input and output. The transistor can be connected in three different circuit configurations with its name relating to the terminal which is common to both the input and output.

These three circuit configurations are the **Common Base** circuit, the **Common Emitter** circuit, and the **Common Collector** circuit as shown in Figure 2.

FIGURE 2. BIPOLAR TRANSISTOR CONFIGURATIONS



As we will see, in each configuration, one of the transistor's leads serves as a common reference point, and the other two leads serve as input and output connections producing particular advantages in terms of voltage gain, current gain, and input and output impedance for each different configuration.

The generalised characteristics of the different transistor configurations given in the following table:

TABLE 1. GENERALISED CHARACTERISTICS

Characteristics	Common Base	Common Emitter	Common Collector
Input Impedance	Low	Medium	High
Output Impedance	Very High	High	Low
Phase Shift	0°	180°	0°
Voltage Gain	High	Medium	Low
Current Gain	Low	Medium	High
Power Gain	Low	Very High	Medium

Note that each transistor amplifier configuration can be constructed using either **NPN** or **PNP** bipolar transistors. The simplest amplifiers are constructed with a single bipolar transistor as its amplifying device and use a few passive components to establish the required bias conditions. These amplifiers are known as single-stage amplifiers.

4. THE COMMON EMITTER CONFIGURATION

The **Common-Emitter (CE)** amplifier configuration is the most widely used transistor circuit. It is called the common-emitter because the transistors emitter terminal is common to both the input and output signals. The input signal is applied between the base and the emitter, while the output signal appears between the transistors collector and emitter terminals.

Thus the base-emitter junction is forward biased and the base-collector junction is reverse biased allowing the transistor to operate as an amplifying device in its active region. This type of configuration is the most commonly used circuit for transistor based amplifiers and which represents the “normal” method of bipolar transistor connection. NPN and PNP common-emitter configurations are shown in Figure 3.

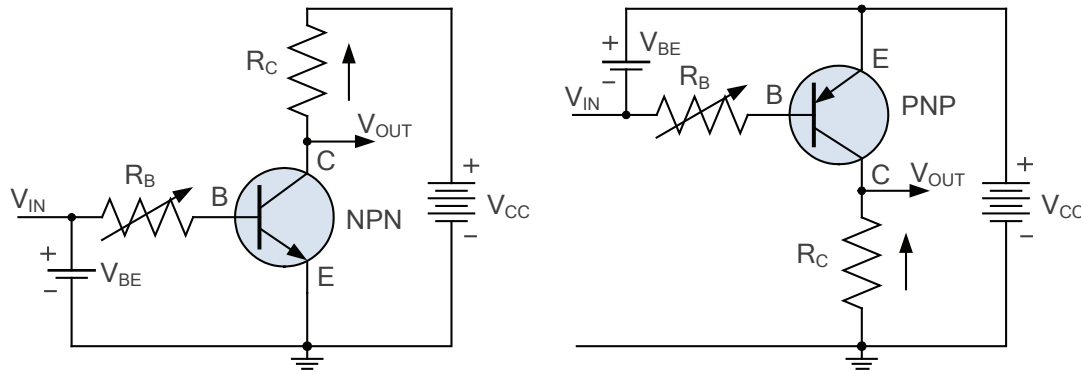
FIGURE 3. COMMON EMITTER CONFIGURATION


Figure 3a. shows the biasing voltages required to turn “ON” an NPN junction transistor, for example a BC237. Figure 3b shows the biasing voltages required for a PNP bipolar junction transistor, for example a BC307. The amplifying action of the PNP transistor is the same as the NPN type, only the bias polarities and directions of current flow are reversed.

For any bipolar transistor, NPN or PNP, the currents flowing out of the transistor must be equal to the currents flowing into the transistor and are related as: $I_e = I_c + I_b$.

As the load resistance (R_C) is connected in series with the collector across the supply, the DC current gain of the common-emitter transistor configuration is quite large as it is the ratio of $\Delta I_c / \Delta I_b$.

A common-emitter transistors DC current gain is given the Greek symbol of **Beta β** . Typical Beta values range between 20 and 200 for most general purpose transistors. Unfortunately, Beta values can also vary a lot between bipolar transistors of the same part/model number due to their manufacturing process.

Since the emitter current for a common-emitter configuration is defined as $I_e = I_c + I_b$, the ratio of I_c / I_e is called Alpha, given the Greek symbol of **α** . Note: that the value of Alpha will always be less than unity.

The Δ (delta) symbol simply means a “change in” the current value

By combining the expressions for both Alpha, **α** and Beta, **β** a mathematical relationship between these parameters and therefore the static DC current gain of the common emitter transistor configuration can be defined as:

$$\text{Alpha, } \alpha = \frac{\Delta I_c}{\Delta I_e} = \frac{\Delta I_c}{\Delta I_c + I_b}$$

$$\text{Beta, } \beta = \frac{\Delta I_c}{\Delta I_b} = \frac{\Delta I_c}{\Delta I_e - I_c}$$

But collector current $I_c = \alpha * I_e = \beta * I_b$

Therefore Alpha and beta are related to each other as:

$$\beta = \frac{\alpha I_e}{I_e - \alpha I_e} = \frac{\alpha}{1 - \alpha} \quad \text{and} \quad \alpha = \frac{\beta I_b}{\beta I_b + I_b} = \frac{\beta}{\beta + 1}$$

The electrical relationship between these three currents, I_b , I_c and I_e is determined by the physical construction of the transistor itself, so any small changes in base current (I_b), will result in a much larger change in collector current (I_c). That is, small changes in current flowing in the base can control a larger current in the emitter-collector circuit.

The voltage gain of the common emitter configuration is given by:

$$A_v = \beta \times R_C / R_B$$

Therefore, to achieve high voltage amplification, transistors with high **β** values should be used and the input base resistance **R_B** of the transistor should be low compared to the collector load resistance **R_C** .

The common emitter configuration is an inverting amplifier circuit. This means that the resulting output signal has a 180° phase-shift with regards to the input voltage signal.

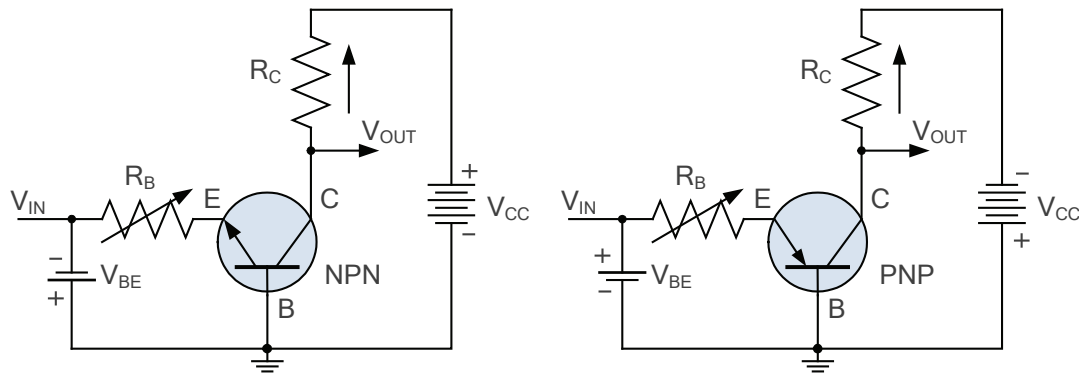
5. THE COMMON BASE CONFIGURATION

As its name suggests, in the **Common Base** (CB) or grounded base configuration, the input signal is applied between the transistors emitter and base terminals while the corresponding output signal is taken from across the collector and base terminals. Thus the base terminal of the transistor is common to both input and output signals. Also, the base terminal can be either grounded or connected to some fixed reference voltage point.

The input current flowing into the emitter is the sum of both the base current and collector current respectively. Therefore, the collector current output is just less than the input emitter current resulting in a current gain for this type of circuit of less than unity.

Then the common-base configuration cannot be used as a current amplifier as it “attenuates” the input signal, but can be used as a voltage amplifier.

FIGURE 4. COMMON BASE CONFIGURATION



Both the NPN and PNP common-base configurations of Figure 4. produce a non-inverting voltage output signal. That is the signal voltages of V_{IN} and V_{OUT} are “in-phase”.

It also has a high ratio of output to input resistance as its input resistance (R_b) is low, while its output resistance (R_c) is high. Thus by selecting proper values for R_b and R_c , substantial voltage gain can be obtained.

We recall from the *common-emitter* configuration that $I_c = \beta I_b$ relating the input base current (I_b) to the output collector current (I_c). In the common-base configuration, the ratio of the change in collector current to the change in emitter current is known as its current amplification factor, alpha α . Where $I_c = \alpha I_e$.

Then the voltage gain, (A_V) for a common-base configuration is given as:

$$\text{Common Base Voltage Gain: } A_V = \frac{V_{OUT}}{V_{IN}} = \frac{I_c \times R_c}{I_e \times R_b} = \alpha \frac{R_c}{R_b}$$

As alpha, and therefore the current gain I_c/I_e is slightly less than unity, it is therefore more useful to express the voltage gain of the common-base as the ratio of the collector load resistance to that of the emitter input resistance. That is: $A_V = R_c/R_b$ (the resistance gain).

Thus if we had the collector load resistance of 40k Ω and input resistance of 40 Ω , then the voltage gain would be: $V_{OUT}/V_{IN} = 250$.

Since the common-base configuration has a very low input resistance, a very high output resistance, high voltage gain and unity current gain. This type of circuit is generally used as a cascode stage in single stage amplifier circuits such as microphone pre-amplifier or radio frequency amplifiers due to its very good high frequency response.

6. THE COMMON COLLECTOR CONFIGURATION

In the Common Collector (CC) configuration, the collector terminal is connected to ground or earth potential. When using an NPN transistor this is achieved through the power supply. For the PNP transistor, the collector is connected directly to ground as shown in Figure 5.

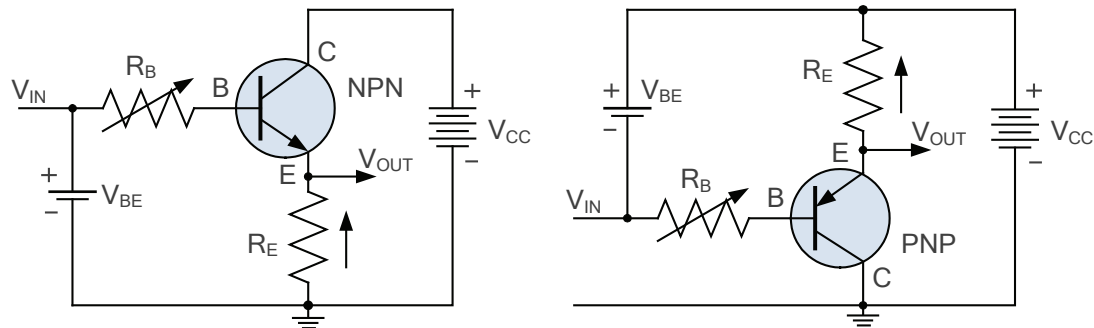
The input signal is applied between the base and collector terminals, the output signal being generated between emitter and collector. Thus the transistors base terminal serves as the input, the emitter terminal is the output, and the collector terminal is common to both the input and output. Hence its name.

This type of configuration is commonly known as a Voltage Follower or Emitter Follower circuit since it displays a high input and low output impedance.

The common-collector, or emitter follower configuration is very useful for impedance matching applications where a high impedance input source needs to be connected to a low impedance output load requiring a high current gain.

In some ways, the *common-collector* configuration is the opposite of the common-emitter configuration, as the connected load resistor is moved from the usual collector terminal to the emitter terminal where it is labelled R_E for emitter resistance as shown in Figure 5.

FIGURE 5. COMMON COLLECTOR CONFIGURATION



The current gain for the common-collector configuration is approximately equal to the beta (β) value of the transistor itself. The emitter current is the combination of the collector current (I_C) and the base current (I_B) combined as: $I_E = I_C + I_B$. Thus only the common-collector configuration has both the input and output currents passing through the same resistor.

The common-collector transistor configuration is used as a current amplifier with the current gain, A_i given as:

$$A_i = \frac{I_e}{I_b} = \frac{I_e}{I_e - I_c} = \frac{I_e}{I_e - \alpha I_e} = \frac{I_e}{(1 - \alpha) I_e} = \frac{1}{1 - \alpha} = \beta + 1$$

This type of bipolar transistor configuration is a non-inverting circuit in that the signal voltages of V_{in} and V_{out} are “in-phase”. The common collector configuration has a voltage gain of about “1” (unity gain). Thus it can be considered as a voltage-buffer since the voltage gain is unity.

The load resistance of the common collector transistor receives both the base and collector currents giving a large current gain (as with the common emitter configuration) therefore, providing good current amplification with very little voltage gain.

7. BIPOLAR TRANSISTOR DC CURRENT RELATIONSHIPS

Having looked at the three different types of bipolar transistor configurations, we can now define the various relationships associated with NPN and PNP transistors. Then the principal properties DC current and gains are summarised in the following table, Table 2.

TABLE 2. DC CURRENTS AND GAINS FOR NPN AND PNP TRANSISTORS

$I_E = I_C + I_B$	$I_C = I_E - I_B$	$I_B = I_E - I_C$
$\alpha = I_C / I_E = \beta / (1 + \beta)$	$\beta = I_C / I_B = \alpha / (1 - \alpha)$	$\alpha = I_E / I_B = \beta + 1$
$I_E = I_C / \alpha = I_B (1 + \beta)$	$I_C = \beta I_B = \alpha I_E$	$I_B = I_E / (\beta + 1) = I_E (1 - \alpha)$
$I_E = I_C / \alpha$	$\beta / \alpha \approx I_E / I_B \approx \beta$	$I_B = I_C / \beta$

The voltage gain, A_v of three bipolar transistor configurations are given as:

Common Base: $A_v = V_o / V_i = \alpha R_L / r_e$

Common Emitter: $A_v = V_o / V_i = -R_c / r_e$

Common Collector: $A_v = V_o / V_i \approx 1$

Where r_e is the ac resistance of emitter-base diode given as: $26\text{mV} / I_E$

8. THE COMMON EMITTER AMPLIFIER

One of the main applications of bipolar junction transistors is in signal amplification. We have seen that in the common-emitter configuration, a small current applied to the base terminal of bipolar transistor can control the flow of a much larger collector current supplied by a DC power source.

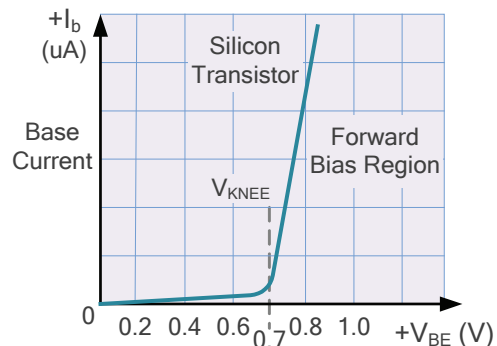
To do this we need to bias the transistor in such a way for it to operate in its active region. The transistor will amplify any small AC signal applied to its base with the emitter grounded. Thus it can be used as an amplifying device.

One such commonly used configuration of an NPN transistor is that of the “Common-emitter Amplifier” configuration, producing what is generally called a “Class-A amplifier” type circuit. But there are many other kinds of BJT amplifier circuit configurations, each with its own special features, advantages and applications.

8.1 Transistor Input Characteristics

The voltage-current (V-I) input characteristics of a bipolar junction transistor is exactly the same as for a forward-biased rectifying diode since the base-emitter junction of the transistor resembles that of a pn-junction as shown in Figure 6.

FIGURE 6. CE INPUT CHARACTERISTIC



Note that the transistors base current, I_b will only flow in the forward direction through this pn-junction when the base-emitter voltage is greater than 0.7 volts for a silicon device, or greater than 0.3 volts for a germanium device.

Also once forward biased, the voltage drop across this base-emitter junction stays relatively constant for varying amounts of base current, I_b . Thus again showing that the bipolar junction transistor is a current operated device.

8.2 Transistor Output Characteristics

The output voltage-current characteristics are a set of graphical curves which show the behaviour of the output collector current, I_c over a wide range of collector-emitter voltages, V_{ce} for a given amount of input base current, I_b drive as shown in Figure 7.

FIGURE 7. COMMON EMITTER OUTPUT CHARACTERISTIC CURVE

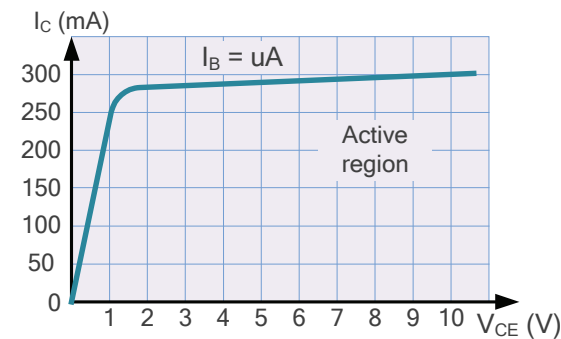


Figure 7. shows the way in which the collector current of a bipolar transistor varies with collector voltage for given values of base current.

By selecting a particular and fixed input base current value, increasing the transistors collector-emitter voltage while measuring the resulting output collector current for a given load resistor, R_c .

Note that the output characteristic line is straight, and virtually horizontal for just one single value of I_b . Therefore, we can generate a whole family of characteristic curves for the same transistor by simply selecting different values of base current from zero amperes ($I_b = 0$) up to the manufactures maximum input value ($I_b = \mu A_{max}$).

You may recall from Table 2. previously, that the DC current gain Beta (β) of the common-emitter configuration is defined as: $\beta = I_c / I_b$.

That is output current divided by input current. So we can use the output characteristics curve to find the static gain of the transistor.

For example, if a base current of $10\mu A$ causes a collector current of 1 mA to flow, then the corresponding value of beta is $I_c / I_b = 1\text{ mA} / 10\mu A = 100$. That is, the transistor has a current amplification factor of 100.

Characteristic curves show the general operation of the common-emitter transistor amplifier circuit

9. BIASING THE CE AMPLIFIER

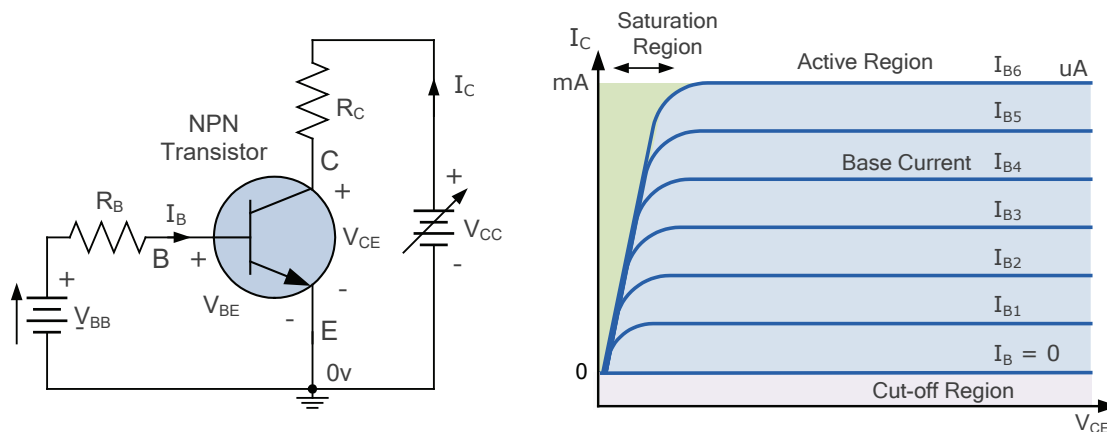
Biasing a bipolar junction transistor means finding the required values of collector-emitter voltage V_{CE} and collector current I_C required to continually forward bias the base-emitter junction allowing current to flow.

Biasing allows the class-A configured amplifier to operate in its active region halfway between its cut-off and saturation regions. That is, when its base-emitter junction is forward-biased, and its base-collector junction is reverse-biased allowing the transistor to function as a variable resistor between its collector and emitter junctions.

When correctly biased, a transistor amplifier will accurately reproduce the positive and negative halves of any AC input signal producing an undistorted but amplified output voltage waveform. Remember that amplification only occurs when the transistor is biased into its active region.

We recall previously that input and output characteristics curves can be plotted for any transistor using a fixed base current value with varying collector-emitter voltage. Thus we can create a full set of output characteristics curves for different values of base current as shown in Figure 8.

FIGURE 8. DETERMINING THE I-V CHARACTERISTICS OF A BJT



Having created a set of characteristics curves for a particular transistor, we can determine its *quiescent operating*, or **Q-point**. This defines its operating region to give an undistorted amplified output by correctly setting the required biasing current and voltage conditions.

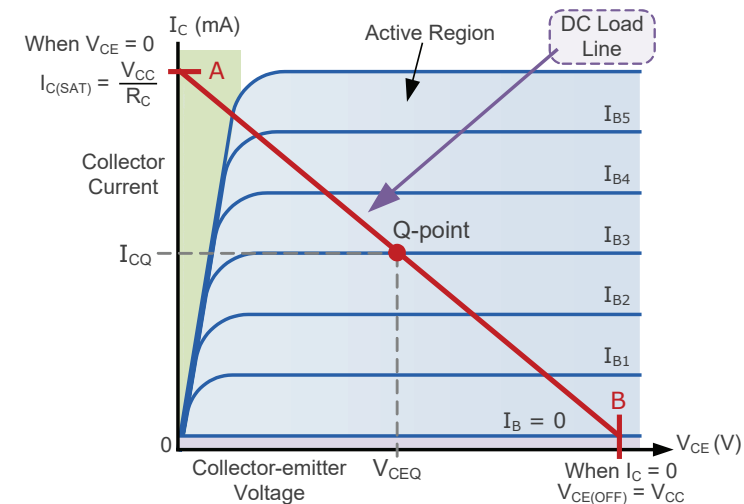
The Q-point can be fixed anywhere within the active region but to ensure that both the positive and negative half cycles of the ac input waveform are correctly amplified. The Q-point is generally centred around the middle of the previous characteristic curves to provide the maximum symmetrical output voltage swing.

10. DC LOAD LINE CONSTRUCTION

When operated as a solid state switch, the transistor is biased between its cut-off and saturation regions. In its cut-off region, $I_C = 0$ and V_{CE} is at maximum as $V_{CE(OFF)} = V_{CC}$. Likewise when it is in its saturation region, I_C is maximum and $V_{CE} = 0.2V$ (zero volts for ideal condition). Collector current I_C is limited only by the collector load resistance, R_C .

A DC load line can be plotted on the characteristic curves between $V_{CE(OFF)} = V_{CC}$ (point "B") on the x-axis, and $I_{C(SAT)} = V_{CC} / R_C$ (point "A") on the y-axis as shown in Figure 9.

FIGURE 9. DC LOAD LINE CONSTRUCTION



Having determined the position of the Q-point at the centre of the load line, the quiescent currents and voltages, I_{CQ} and V_{CEQ} respectively can be found. As the Q-point is centred on the load line between points **A** and **B**, with no ac signal input applied to the base, the quiescent collector current I_{CQ} will be equal to half $I_{C(SAT)}$. Thus V_{CEQ} will be equal to half V_{CC} .

The advantage of centring the Q-point on the DC load line is that it allows for the optimum ac operation of the bipolar transistor class-A amplifier when connected in its common emitter configuration. Thus the relationship of: $I_C = \beta_{DC} * I_B$ is once again valid.

However, the Q-point is greatly affected by changes in a transistors beta value. Changes in operating temperature or the replacement of the transistor for another one even of the same number will cause changes in beta. As a result the Q-point will move along the load line affecting its DC operating point.

Biasing a bipolar transistor with a DC voltage ensures that the base-emitter junction is always conducting.

Biasing the transistors base with a fixed voltage value prevents, or at least greatly limits the movement of the Q-point thereby increasing stability of the CE amplifier circuit.

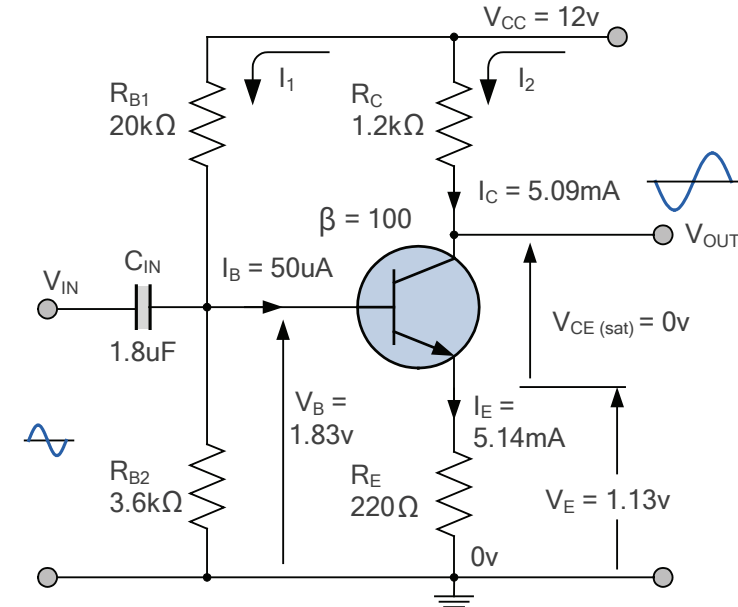
11. BASE BIASING THE TRANSISTOR AMPLIFIER

There are many different ways to bias the base terminal a bipolar junction transistor, such as: *fixed current bias, emitter feedback biasing, collector feedback biasing, dual supply biasing*, etc. But the most common form of biasing the base of a transistor when in class-A mode is by using a voltage divider network.

Consider the common emitter (CE) transistor amplifier of Figure 10. which has a voltage divider biasing network connected to the base consisting of series resistors R_{B1} and R_{B2} , plus an external emitter leg resistance, R_E .

Remember that the common emitter amplifier has a load resistor a part of its collector circuit. The collector current through this resistance (R_C) produces an I^2R voltage output of the amplifier. Thus providing voltage amplification.

FIGURE 10. DC ANALYSIS OF THE AMPLIFIER CIRCUIT



1). As resistors R_{B1} and R_{B2} form a series connected voltage divider network across the supply voltage, V_{CC} . The base biasing voltage V_B is calculated as:

$$V_B = \left(\frac{R_{B2}}{R_{B1} + R_{B2}} \right) \times V_{CC} = \left(\frac{3.6k}{20k + 3.6k} \right) \times 12 = 1.83 \text{ Volts}$$

2). The emitter terminal voltage, V_E

$$V_E = V_B - V_{BE} = 1.83 - 0.7 = 1.13 \text{ Volts}$$

3). The emitter current, I_E

$$I_E = V_E / R_E = 1.13 / 220 = 5.14 \text{ mA}$$

4). Determine the collector (load) current, I_C

$$I_C = \left(\frac{\beta}{\beta + 1} \right) \times I_E = \left(\frac{100}{100 + 1} \right) \times 5.14 \times 10^{-3} = 5.09 \text{ mA}$$

Note that the difference between I_C and I_E will be the base current I_B as: $I_B = I_E - I_C$

5). The quiescent base current, I_B

$$I_B = I_E - I_C = 0.00514 - 0.00509 = 50 \mu\text{A}$$

6). The voltage drop across load resistor, R_C

$$V_{RC} = I_C \times R_C = 0.00509 \times 1200 = 6.1 \text{ Volts}$$

7). The collector-emitter voltage, V_{CE}

$$V_{CE} = V_{CC} - V_{RC} - V_E = 12 - 6.1 - 1.13 = 4.77 \text{ Volts}$$

8). Biased DC output voltage, $V_{OUT(DC)}$

$$V_{OUT} = V_{CC} - V_{RC} = 12 - 6.1 = 5.9 \text{ Volts}$$

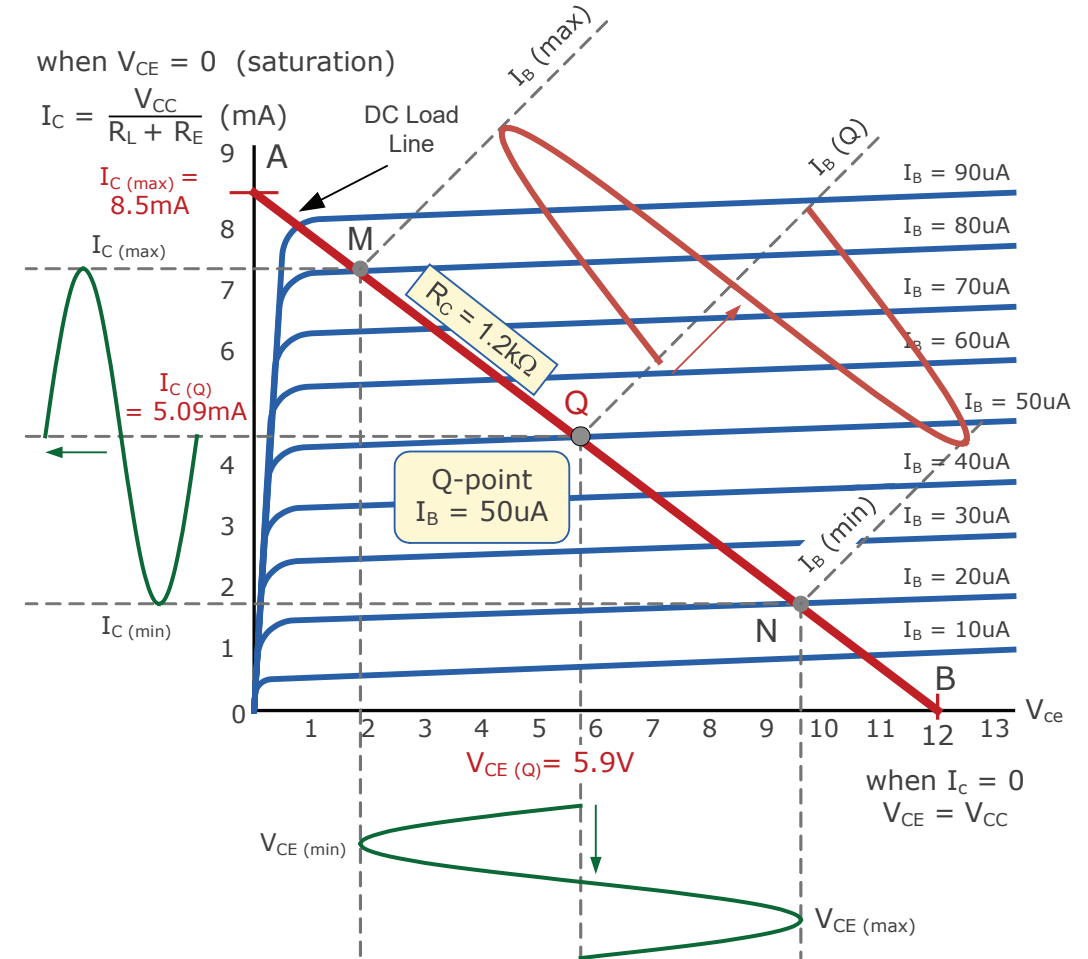
Thus the Q-point sits halfway along the load line and is inside the Active region of the characteristics as shown in Figure 11.

The voltage gain of the transistor amplifier is given as: $A_v = V_{OUT}/V_{IN}$ where $V_{OUT} = I_C \times R_C$ and $V_{IN} = I_E \times R_E$. Since I_E is approximately equal to I_C ($I_E \approx I_C$) the two current values effectively cancel each other out giving the DC voltage gain (excluding r_e) of the circuit as being:

$$A_v = \frac{V_{OUT}}{V_{IN}} \cong \frac{R_C}{R_E} = \frac{1200}{220} = 5.45$$

So for an input signal voltage of $0.5V_{pk}$, the output voltage will be: $5.45 \times 0.5 = 2.725V_{pk}$. Remember that for a common-emitter amplifier, a positive value for the voltage gain implies its output voltage is inverted with respect to the input signal voltage.

FIGURE 11. TRANSISTOR AMPLIFIER CHARACTERISTICS CURVES



Then we can see that by biasing the transistors Q-point half-way along the DC load line, when an input signal is applied, the base current swings between $20 \mu\text{A}$ and $80 \mu\text{A}$. Thus producing a collector output current I_C swing of 7.2 mA and a voltage swing of approximately 8 volts .

12. AMPLIFIER DISTORTION

We have seen that for a small signal amplifier to operate correctly without any amplifier distortion of the output signal, it requires some form of DC Bias on its Base terminal. This allows it to amplify its input signal over the entire cycle with the bias “Q-point” set as near to the middle of the load line as possible. Thus the bias Q-point setting for the common-emitter amplifier will give us a “Class-A” type amplification configuration.

However, if we incorrectly design our amplifier circuit and set the biasing Q-point at the wrong position on the load line or apply too large an input signal to the amplifier, the resultant output waveform may not be an exact reproduction of the original input signal waveform. In other words the amplifiers output signal will suffer from what is commonly called **Amplifier Distortion**.

Amplifier Distortion as a result of *amplitude distortion* occurs when the peak values of the output waveform are attenuated causing distortion. This is mainly due to a shift in the Q-point along the load line preventing amplification to occur over the whole signal cycle.

FIGURE 12. AMPLITUDE DISTORTION DUE TO INCORRECT BIASING

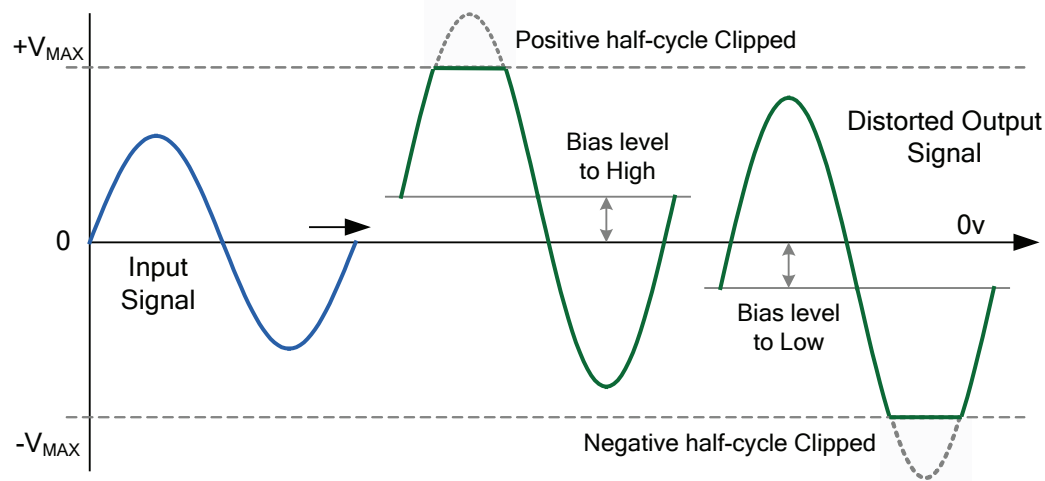


Figure 12. shows that if there is insufficient bias, the Q-point lies in the lower half of the load line so the output waveform will look like the one on the right with the negative half of the output waveform “cut-off” or clipped. Likewise, if the bias level is too high and the Q-point lies in the upper half of the load line, then the output waveform will look like the one on the left with the positive half “cut-off” or clipped.

Even with correct Q-point biasing, it is still possible for the output voltage waveform to become distorted due to a large input signal being amplified by the circuits voltage gain. As a result, the output voltage signal becomes clipped during both the positive and negative parts of the waveform and no longer resembles the input signal.

This type of amplitude distortion is called *clipping* as shown in Figure 13. It is commonly the result of “over-driving” the input of the amplifier.

FIGURE 13. AMPLITUDE DISTORTION DUE TO CLIPPING

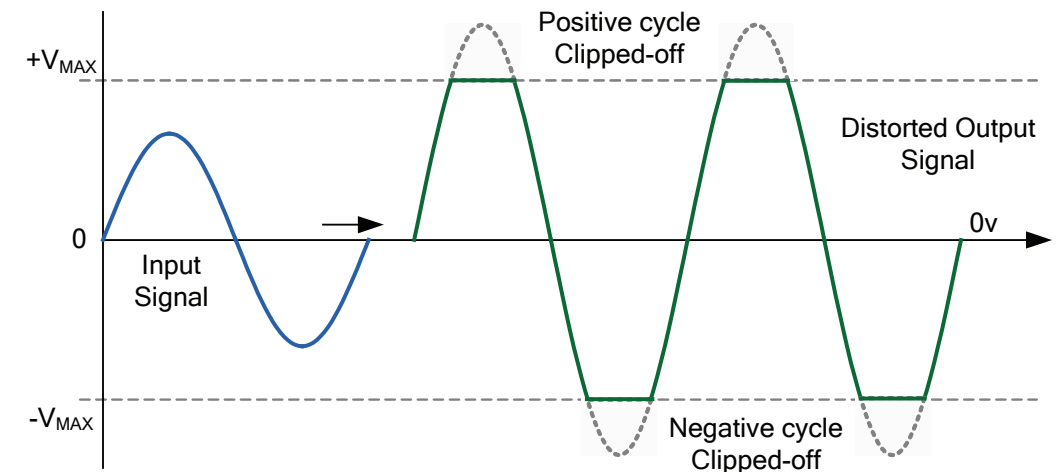


Figure 13. shows the effect when the amplitude of the input signal is too large. Clipping of the output waveform forces it to exceed the power supply voltage rails with the peak +ve and -ve halves of the output waveform becoming flattened or “Clipped-off”.

Amplitude distortion greatly reduces the efficiency of a small signal amplifier circuit. Also, the “flat tops” of the distorted output waveform either due to incorrect biasing or over driving the input do not contribute anything to the strength of the output signal at the desired frequency.

However, some well known guitarist and rock bands actually prefer that their distinctive sound is highly distorted or “overdriven” by heavily clipping the output waveform to both the +ve and -ve power supply rails.

Also, increasing the amounts of clipping on an output waveform can produce so much amplifier distortion that it will eventually produce an output waveshape which resembles that of a “square wave” which could then be used in electronic synthesizer circuits.

[End of this Transistor Amplifier eBook](#)

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Main Headquarters
245 Main Street
Cambridge, MA 02142
www.aspendore.com

Central Europe/EMEA
Frankfurter Strasse 211
63263 Neu-Isenburg, Germany
info-europe@aspencore.com